Rack plate classifier: clean vs dirty

DishMATE · NEURA MAiRA dishwashing workcell · 20 October 2026

Summary. The original plate classifier was trained on top-down photos of plates on a white table. In the robot's rack setup it was right on only 16 of 63 photos, worse than always answering "dirty". We replaced its decision step with a small new last layer trained on photos from the rack setup, on top of ImageNet MobileNetV2 features of the cropped plate area. Tested on capture days it had not seen, the new classifier gets **61 of 63** right and **never calls a dirty plate clean**. It is now the default classifier for the robot.

25 %	97 %	0 / 41	2 / 22
original CNN, rack photos	new classifier, unseen days	dirty plates called clean	clean plates called dirty

1. The problem

The robot photographs each plate from its observe pose (GP156) and sorts it: clean plates to the table (GP164), dirty plates to the dishwasher rack (GP166, GP167). In practice the classifier often got this wrong. For example, the clearly stained plate in 20 Oct `plate_5` scored 0.04 ("very clean"), and the clean plate in `plate_6` scored 0.50 ("dirty").

The cause is a **domain gap**. The original model learned from top-down photos of a single plate on a white table. The robot sees the plate side-on in a grey dish rack, with towels, rack bars and often a second plate behind it. Most of the picture is background the model has never seen, and its scores on rack photos are close to random.

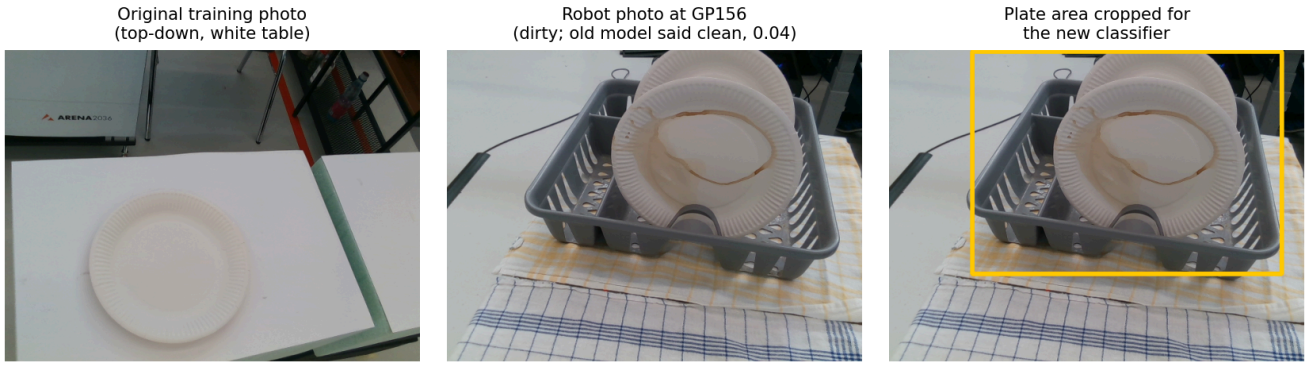


Figure 1. Left: the kind of photo the original model was trained on. Middle: what the robot sees at GP156. Right: the fixed plate area (yellow) the new classifier looks at.

2. Models

Original model (before)

MobileNetV2 convolutional network, global average pooling, dropout 0.5, and one sigmoid output ("dirty probability"). Input: the full camera image resized to 224 × 224 pixels. All layers were trained on our table-top photos.

New rack classifier (now)

The idea, suggested in the project, was to stop relying on the final decision of the old network and instead compare the *features* of a new photo with those of known clean and dirty plates. A MobileNetV2 network turns an image into 1,280 numbers just before its last layer; similar-looking plates give similar numbers. The final design:

1 · Crop
Fixed plate area of the GP156 photo: 20–95 % of the width, top 72 % of the height.

→

2 · Features
MobileNetV2 pretrained on ImageNet (about 1.3 million general photos), unchanged. Output: 1,280 numbers, scaled to length 1.

→

3 · New last layer
Logistic regression: 1,280 weights + 1 bias, trained on our labelled rack photos.

→

4 · Decision
Dirty probability above 0.5 → dirty (GP166/GP167), otherwise clean (GP164).

Only step 3 is trained on our data, so it learns from very few photos without overfitting the large network. Training uses gradient descent (3,000 steps, learning rate 0.5, L2 regularisation 0.001). Clean and dirty photos are weighted equally, because the data has about twice as many dirty plates. Training takes about a minute on the CPU. Code: `rack_classifier.py` ; trained layer: `models/plate_rack_head.npz` .

3. How the approaches were compared

Data. 63 photos taken by the robot at GP156 in the rack setup over four days (9, 12, 16 and 20 October), 41 dirty and 22 clean. Each photo was labelled by the condition of the front, fully visible plate, the one the robot is judging. The labels are listed in `datasets/rack_labels.csv` .

Leave-one-day-out testing. Photos from the same day often show the same plates several times, so testing on photos the model was trained on would look far better than reality. Each day was therefore predicted by a model built only from the other three days, and the results were combined. All rack numbers below are measured this way.

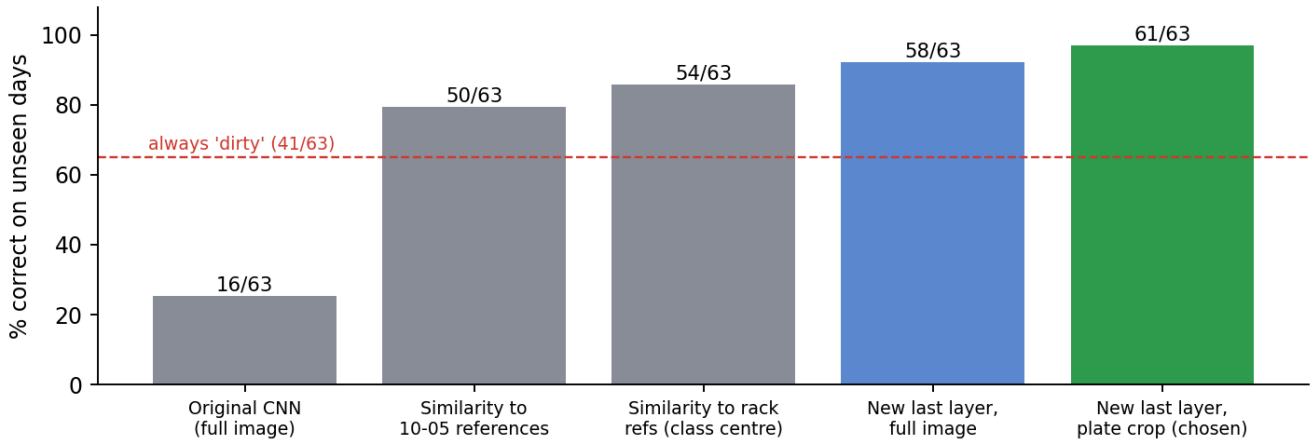


Figure 2. Correct predictions on unseen days for each approach tried. The dashed line is the trivial "always dirty" answer (41/63).

Approach	Correct	Finding
Original CNN, full image	16 / 63	Fails on the rack setup.
Feature similarity to the 10-05 reference plates (3 clean, 2 dirty)	best 50 / 63	The references are table-top photos; similarity is dominated by the background, not the stains. Still calls `plate_5` clean.
Similarity to rack-photo references (nearest class average)	best 54 / 63	References from the same setup work much better.
New last layer on ImageNet features, full image	58 / 63	Learning which features matter beats plain similarity.
New last layer on ImageNet features, plate area cropped	**61 / 63**	**Chosen.** Cropping removes most of the towels and table.

Features from the original CNN were also tried in place of the ImageNet features; they were 2–9 photos worse in each setting. ImageNet features describe general textures and colours, which carry over better to the new scene.

4. Results of the chosen classifier

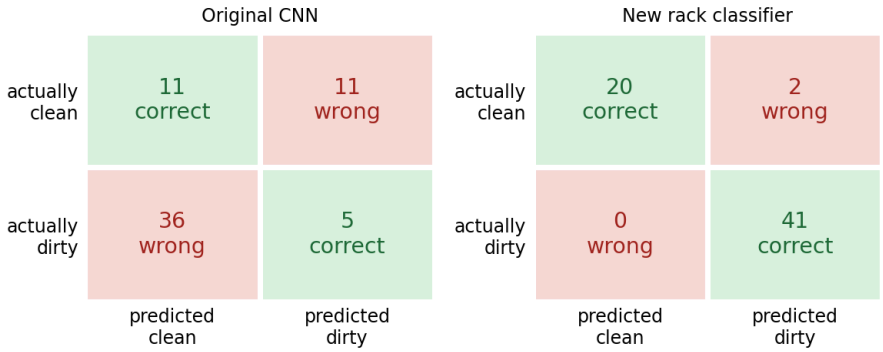


Figure 3. Predictions on all 63 rack photos (unseen days for the new classifier).

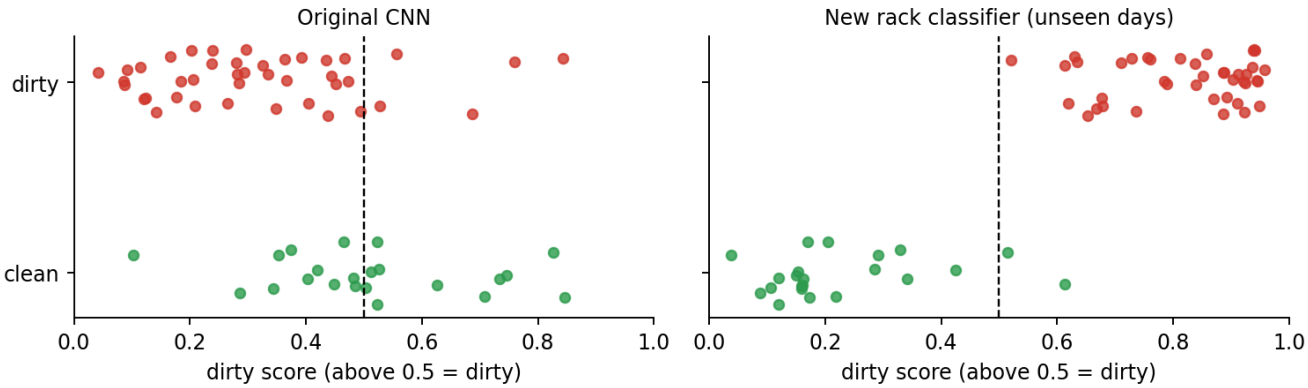


Figure 4. Dirty score of every photo. The old model's scores for clean and dirty plates overlap completely; the new classifier separates them, with most photos far from the 0.5 threshold.

Capture day	Photos	Dirty	Original CNN	New classifier
9 Oct	29	24	14 %	100 %
12 Oct	14	7	43 %	100 %
16 Oct	14	7	29 %	86 %
20 Oct	6	3	33 %	100 %
All	**63**	**41**	**25 %**	**97 %**

The two photos from the problem report, scored with 20 October left out of training: `plate_5` (dirty) now scores **0.89** (was 0.04), and `plate_6` (clean) **0.11** (was 0.50). Both are correct.

The two remaining errors are clean plates called dirty, both from 16 October with a blue bowl standing behind the plate, a scene that appears nowhere else in the data. Erring this way only sends a clean plate to the dishwasher rack; no dirty plate was sent to the clean stack.

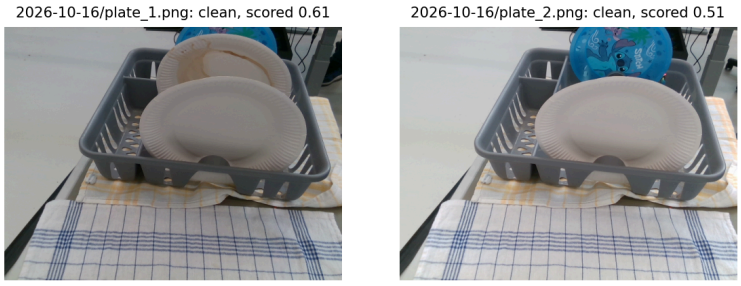


Figure 5. The two photos the new classifier gets wrong on unseen days.

5. Limitations

- **Small data set.** 63 photos from four days, with repeated plates. The 97 % is an estimate from this data and will vary on new scenes (other objects in the rack, different light, other plate types).
- **Labels by eye.** Photos were labelled by the front plate. A stained plate behind a clean front plate counts as clean.
- **Fixed crop.** The crop assumes the rack and the GP156 camera pose stay where they are. If either moves, the crop in `rack_classifier.py` (`ROI`) must be adjusted and the layer retrained.
- **Specific to the rack setup.** Top-down table-top photos are not what it was trained for; the original CNN is kept for those.

6. Using and improving it

The robot uses the new classifier automatically, through `classify_image.py` . The original CNN is still available with `classify_image.py --cnn IMAGE` . To improve the classifier, add photos it got wrong to `datasets/rack_labels.csv` (one line per photo, e.g. `captures/2026-10-21/plate_3.png,dirty,`) and retrain in the classifier environment:

```
TF_CPP_MIN_LOG_LEVEL=3 /root/.venvs/dishwashing-classifier/bin/python rack_classifier.py train
```

This prints the leave-one-day-out accuracy and the photos still misclassified, then saves the new layer. The next plate the robot photographs uses it. Recommended next data: clean plates with other objects behind them (the blue-bowl case), more clean plates in general (the data is two-thirds dirty), and photos under different lighting.